

## CASE STUDY

# NYC CITI BIKE DASHBOARD (2022)

### TOOLS & LIBRARIES

Python · Pandas · NumPy · Plotly · Seaborn · Matplotlib · Streamlit · Kepler.gl

**Data source:** NYC Citi Bike system data (2022) — 12 monthly ZIP files, ~29.8 million individual trip records

**Weather data:** NOAA GHCND Daily Summaries, station USW00014732 (LaGuardia Airport) — retrieved via API (TAVG, TMIN, TMAX, PRCP)

**Output:** Fully deployed multi-page Streamlit dashboard covering seasonal trends, temperature correlation, station demand ranking, and geospatial trip clusters



### SOURCE CODE

[github.com/ageelalramadhan/citi-bike-weather-2022](https://github.com/ageelalramadhan/citi-bike-weather-2022) →

5

NOTEBOOKS

29.8M

TRIP RECORDS

364

DAYS MERGED

6

DASHBOARD  
PAGES

## PROJECT OVERVIEW

Citi Bike generates millions of trip records every year, but raw trip logs don't tell operations teams when to scale supply, which stations to prioritize, or how to plan for seasonal swings. This project takes 29.8 million individual ride records, merges them with NOAA daily weather data from LaGuardia Airport, and turns the combined dataset into an interactive, deployed dashboard built for operational decision-making.

**Business question:** How should Citi Bike scale bike supply and rebalancing operations across the year — and which stations, seasons, and environmental factors drive that decision?

## DATA PIPELINE

| STEP                          | SOURCE                      | ACTION                                                 |
|-------------------------------|-----------------------------|--------------------------------------------------------|
| 1. <a href="#">Acquire</a>    | citi_bike_weather_2022      | Downloaded 12 monthly Citi Bike ZIP files + NOAA A     |
| 2. <a href="#">Extract</a>    | citi_bike_weather_2022      | Extracted ZIPs with Python's zipfile module; combin    |
| 3. <a href="#">Clean</a>      | citi_bike_weather_2022      | Parsed timestamps, filtered to 2022-only, dropped r    |
| 4. <a href="#">Aggregate</a>  | citi_bike_weather_2022      | Grouped 29.8M trip rows (2022 only) to daily ride cc   |
| 5. <a href="#">Merge</a>      | citi_bike_weather_2022      | Inner join of daily ride counts with NOAA weather or   |
| 6. <a href="#">Visualize</a>  | exercise_2_3 / exercise_2_4 | Matplotlib and Seaborn EDA: time series, dual-axis c   |
| 7. <a href="#">Geospatial</a> | exercise_2_5                | Kepler.gl trip origin map — spatial cluster analysis o |
| 8. <a href="#">Dashboard</a>  | exercise_2_6                | 7-page Streamlit app with Plotly charts, sidebar navi  |
| 9. <a href="#">Deploy</a>     | app_part_2.py               | Deployed to Streamlit Cloud — live at the link below   |

## SEASONALITY DRIVES DEMAND

Average daily rides follow a clear seasonal arc across 2022. Ridership climbs from a January low of around **33,000 rides/day**, peaks at roughly **115,000 rides/day** in late summer, then falls sharply as temperatures drop. Summer averages **112,153 rides/day** against a winter average of **57,086** — almost a 2× spread.

Average Daily Rides per Month — 2022

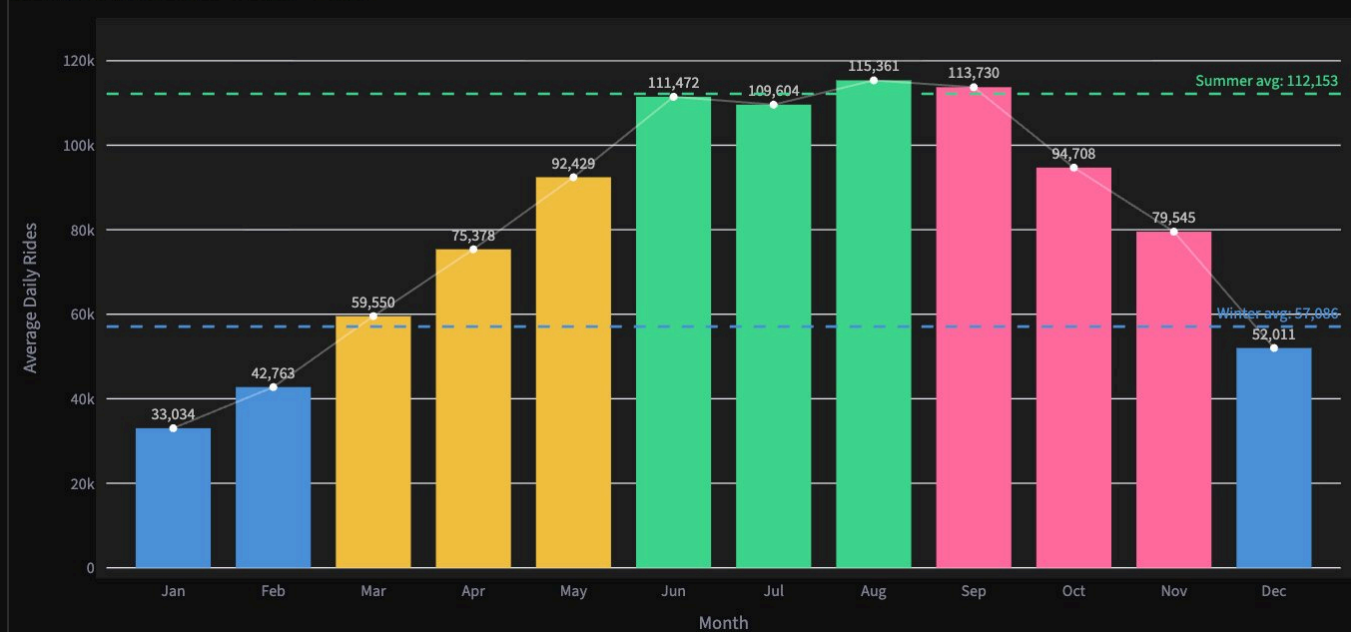


Figure 1. Average daily rides per month, 2022. Bars are coloured by season; reference lines mark the summer (~112K) and winter (~57K) daily averages.

### Key Insight — Seasonal scaling baseline

Winter demand (Nov–Apr) averages roughly half of summer demand (Jun–Aug). This directly quantifies how much redistribution intensity and staffing can be reduced in winter — and how early the summer ramp-up should begin.

## TEMPERATURE PREDICTS DAILY RIDERSHIP

Overlaying daily ride counts with NOAA average temperature shows the two signals tracking each other day-by-day across the full year. The scatter view turns the same relationship into a single trend line: each additional degree of average temperature corresponds to a measurable lift in expected ridership.

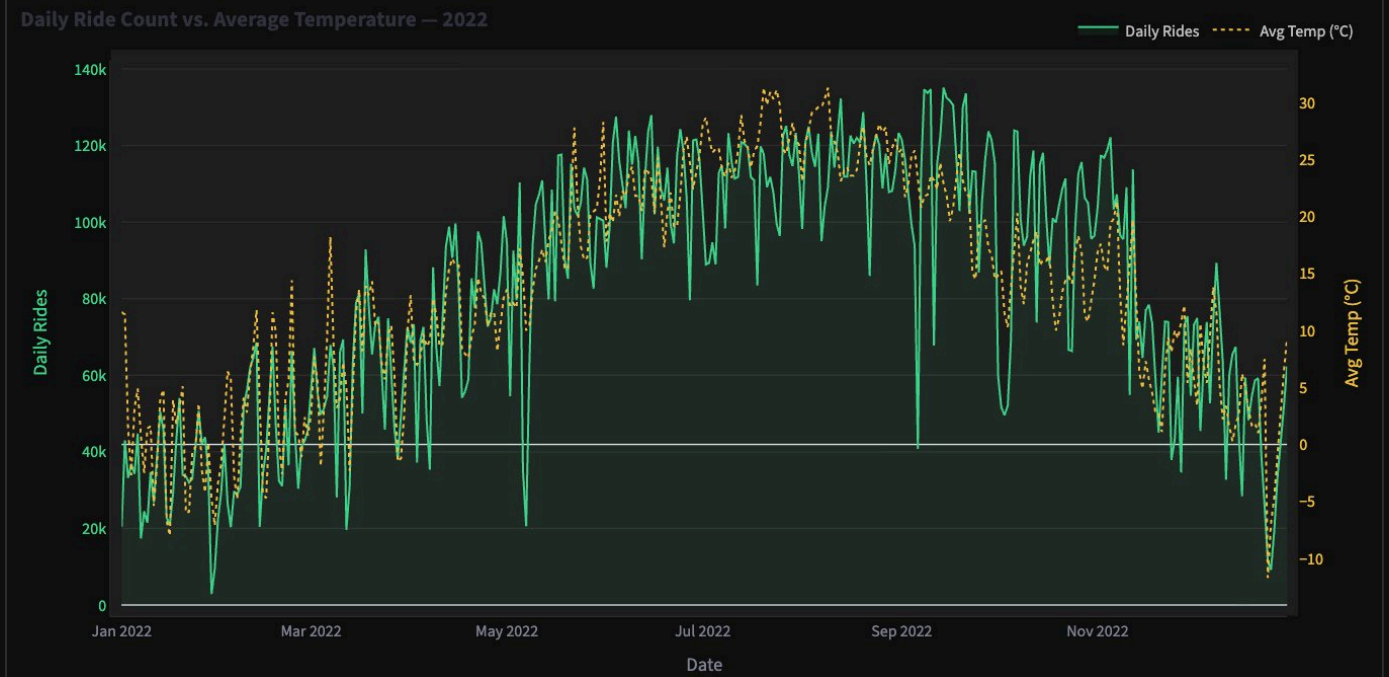


Figure 2. Dual-axis daily series: rides (green) vs average temperature (orange dashed). The two curves move in lockstep across the entire year.

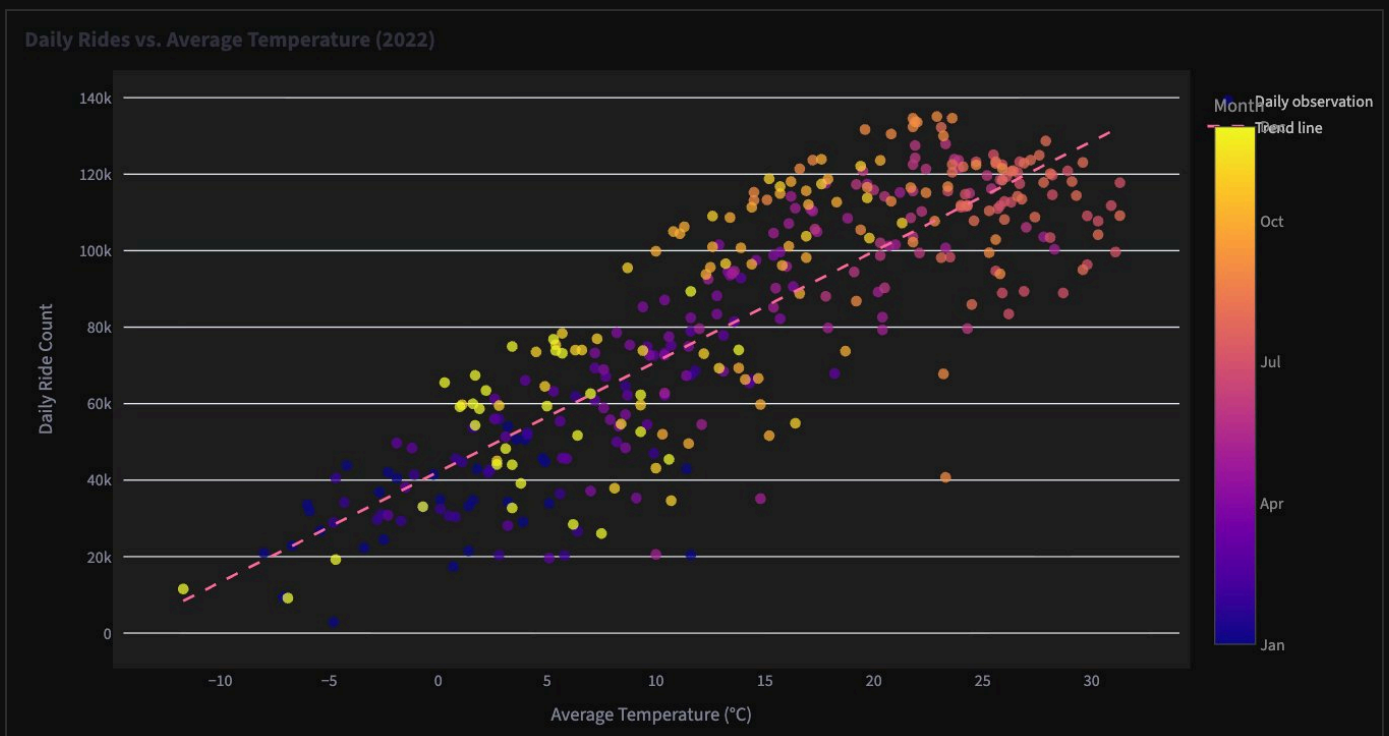


Figure 3. Scatter view of the same relationship, coloured by month. The trend line is positive and roughly linear across the comfortable riding range; cold winter days cluster in the bottom-left, peak summer in the top-right.

### Key Insight — Temperature as a planning signal

Daily ridership tracks closely with daily average temperature. Operations should use forecast temperature (not just calendar month) as a proximal signal for next-day demand scaling — particularly during shoulder seasons.

## STATION CONCENTRATION: WHERE DEMAND SITS

Trip demand is heavily concentrated. The single busiest start station — W 21 St & 6 Ave — generated **128,436 trips** in 2022. The rest of the top 10 sit in a tight band between 94K and 122K, all within central Manhattan: from Chelsea and Tribeca up through Midtown to the Upper East Side.

**Top 10 Most Popular Citi Bike Start Stations (2022)**

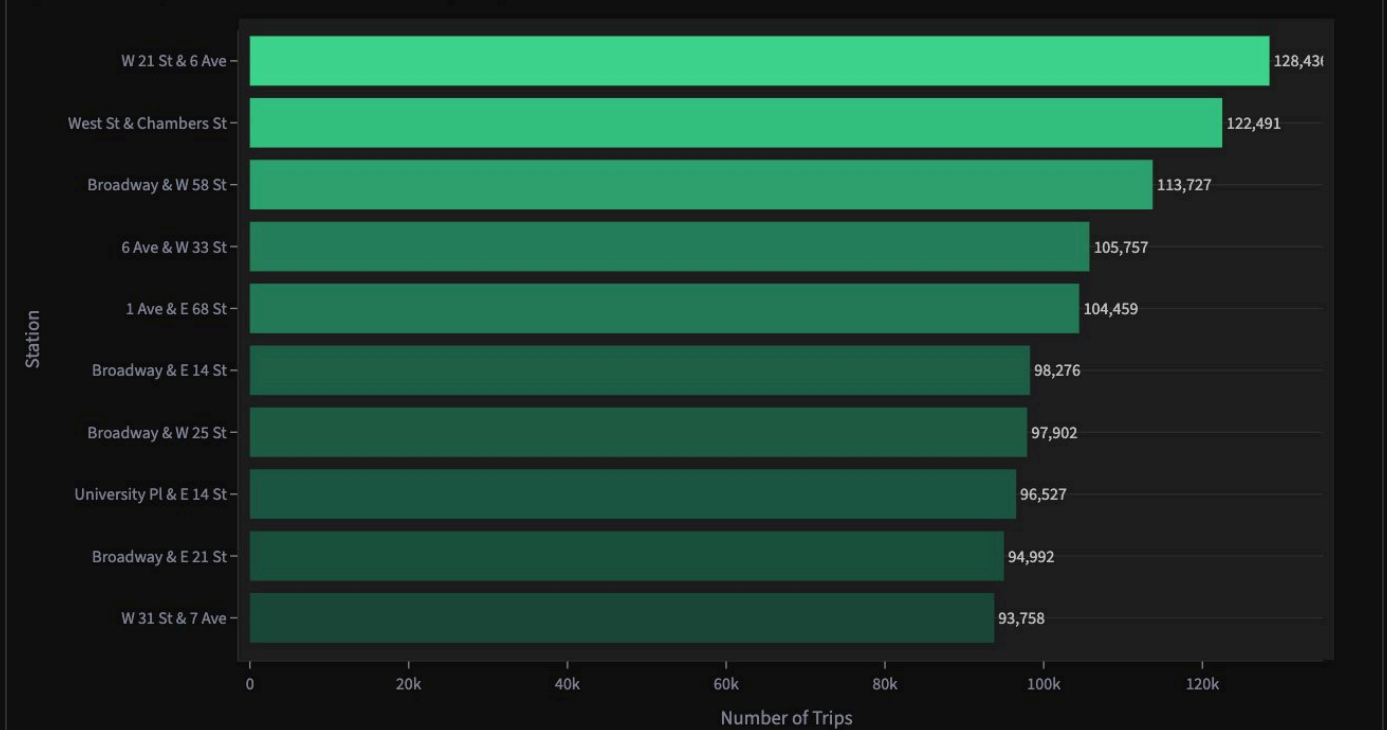


Figure 4. Top 10 most popular start stations. All ten sit in central Manhattan — making them high-priority targets for dock capacity and rebalancing frequency.

### Key Insight — Rebalancing leverage is concentrated

Because trip starts cluster heavily at a small set of central Manhattan stations, rebalancing crews can drive disproportionate operational gains by prioritising redistribution to and from these top 10 stations rather than allocating attention uniformly across the network.

## SPATIAL TRIP PATTERNS

The Kepler.gl map reveals movement patterns invisible in station-level aggregates. Trip clusters are concentrated in Manhattan's business corridor and along the Hudson waterfront, with secondary clusters in Lower Manhattan and Brooklyn. The arcs show where riders actually go — not just where they start.

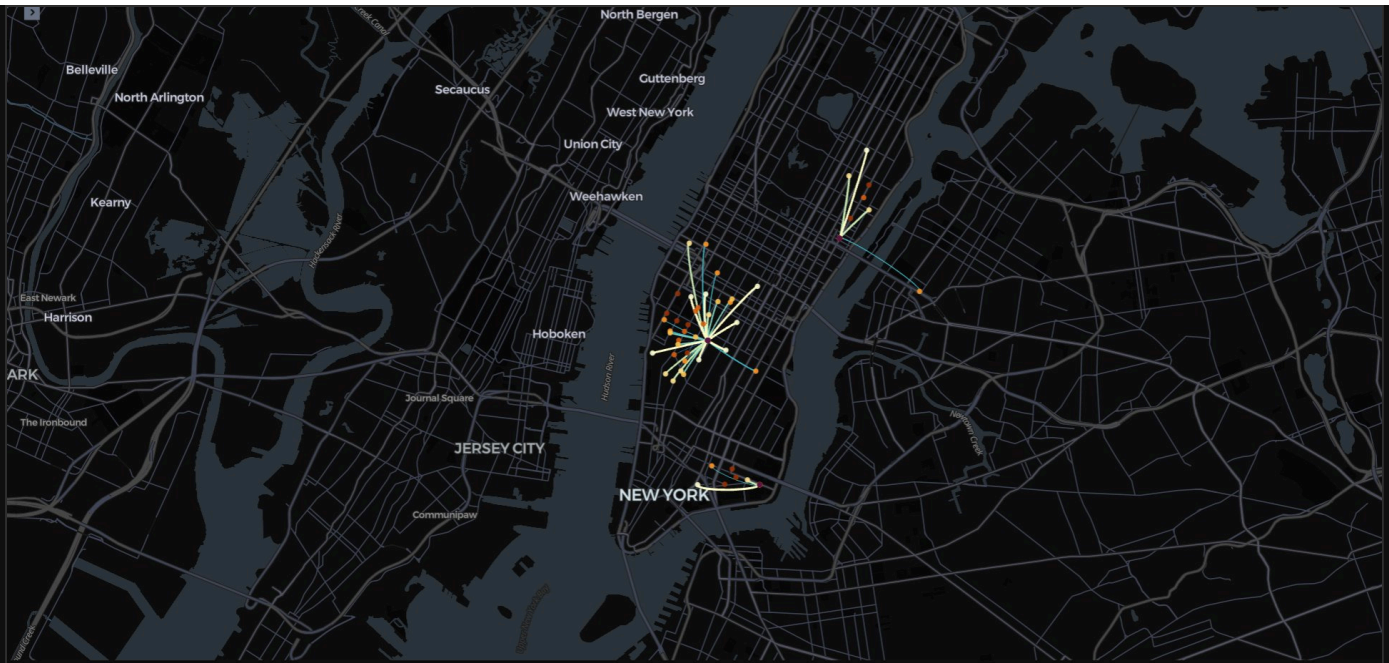


Figure 5. Kepler.gl spatial map: trip origins, destinations, and arcs across NYC. Density confirms the Manhattan-centric pattern seen in the station rankings, and surfaces secondary hubs that the bar chart alone hides.

## KEY FINDINGS

### Finding 1 — Demand triples between winter and summer.

Daily rides scale ~3x from January (~33K) to August (~115K). Winter operations should run at roughly half of summer intensity, with a deliberate ramp from April onward.

### Finding 2 — Temperature is a stronger planning signal than calendar month.

Forecast temperature predicts next-day ridership more precisely than month alone. Operational scaling should use temperature forecasts as a proximal demand input — especially during shoulder seasons.

### Finding 3 — Demand is concentrated, so rebalancing leverage is concentrated.

The top 10 stations capture a disproportionate share of trip starts. Prioritizing redistribution at these stations yields higher operational ROI than uniform attention.

### Finding 4 — A deployed dashboard converts ad-hoc analysis into recurring decisions.

Moving from one-off charts to a multi-page Streamlit app gives operations teams self-service access to seasonal, weather, and station insights — making the analysis directly actionable.

## RESOURCES

[View Live Streamlit Dashboard →](#)

[</> GitHub Repository](#)